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Lorenzo Pompili

Experience

2025–2028 **Research Fellow**
School of Mathematical Sciences, University of Nottingham, Nottingham (United Kingdom)
Nottingham Centre of Gravity, group of Dr. Stephen Green

Education

2021–2025 **PhD in Theoretical Physics, summa cum laude**
Max Planck Institute for Gravitational Physics, Potsdam (Germany)
PhD thesis: *Ever more accurate effective-one-body waveforms for gravitational-wave astrophysics*. Supervisor: Prof. Alessandra Buonanno (Max Planck Institute for Gravitational Physics)

2018–2021 **Master's Degree in Theoretical Physics, 110/110 cum laude**
Università degli Studi di Perugia, Perugia (Italy)
Master's thesis: *Near-extreme Kerr magnetospheres*. Supervisor: Prof. Marta Orselli (Università degli Studi di Perugia, Niels Bohr Institute)

2015–2018 **Bachelor's Degree in Physics, 110/110 cum laude**
Università degli Studi di Perugia, Perugia (Italy)
Bachelor's thesis: *Relativistic hydrodynamics*. Supervisor: Prof. Gianluca Grignani (Università degli Studi di Perugia)

Collaborations and research networks

2025– GWSky member (ERC Synergy Grant *Making Sense of the Unexpected in the Gravitational-Wave Sky*, gwsky.org)

2024–2026 LISA Distributed Data Processing Center member

2023– LISA Consortium core member

2022– Einstein Telescope Collaboration member

Coordinator of Division 8 (Waveforms) of the Observational Science Board (2026–)

2021– LIGO Scientific Collaboration member

Active in the Waveforms, Parameter Estimation and Testing General Relativity working groups, including liaison and coordination roles, technical reviews, and membership of paper editorial teams and dedicated event task forces

Awards

03/2025 Poster prize winner, LIGO-Virgo-KAGRA Collaboration Meeting, Melbourne – *Constraints on quasi-normal-mode frequencies from GWTC-4 observations*

2016–2021 A.Di.S.U. scholarship, awarded consecutively for academic merit during Bachelor's and Master's programs, Università degli Studi di Perugia

Publications (short author-list)

Publications of which I am lead or co-lead author are marked with *.
Publications with author-list in alphabetical order are marked with **.

[21] Cheng Foo, Alessandra Buonanno, Hector Estelles, **Lorenzo Pompili**, *Pushing the accuracy of the quasi-circular baseline for the SEOBNR family of models*

Submitted to Phys. Rev. D, arXiv:2611.11449

[20] Aldo Gamboa, **Lorenzo Pompili**, Alessandra Buonanno et al., *Enabling gravitational-wave astronomy with spin-precessing black holes on generic orbits*

Submitted to Phys. Rev. Lett., arXiv:2609.01568

[19] * **Lorenzo Pompili**, Aldo Gamboa, Alessandra Buonanno, *Joint inference of line-of-sight acceleration and orbital eccentricity in neutron star black hole binaries*

Submitted to Phys. Rev. D, arXiv:2606.28156

[18] * **Lorenzo Pompili**, Aldo Gamboa, Alessandra Buonanno, *Eccentric and unbound compact binaries in the LIGO-Virgo-KAGRA catalog: parameter estimation and waveform systematics with SEOBNRv6EHM*

Phys. Rev. D 114 (2026) 10, 104009, arXiv:2605.28716

[17] Aldo Gamboa, Alessandra Buonanno, **Lorenzo Pompili** et al., *Accurate waveforms for generic planar-orbit binary black holes: The multipolar effective-one-body model SEOBNRv6EHM*

Phys. Rev. D 114 (2026) 10, 104010, arXiv:2605.28715

[16] Nihar Gupte et al. (inc. **Lorenzo Pompili**), *Eccentricity constraints disfavor single-single capture in nuclear star clusters as the origin of all LIGO-Virgo-KAGRA binary black holes*

Astrophys. J. 1009 (2026) 1, 90, arXiv:2603.29019

[15] Leonardo Grimaldi, Elisa Maggio, **Lorenzo Pompili**, Alessandra Buonanno, *Plunge–Merger–Ringdown Tests of General Relativity with GW250114*

Phys. Rev. D (*Letter*) 113 (2026) 6, L061506, arXiv:2601.13173

[14] Hector Estelles, Alessandra Buonanno, Raffi Enficiaud, Cheng Foo, **Lorenzo Pompili**, *Incorporating antisymmetric multipoles for spin-precessing binary signals into the SEOBNR framework*

Phys. Rev. D 113 (2026) 4, 044049, arXiv:2506.19911

[13] * **Lorenzo Pompili**, Elisa Maggio, Hector O. Silva, Alessandra Buonanno, *A parametrized spin-precessing inspiral-merger-ringdown waveform model for tests of general relativity*

Phys. Rev. D 111 (2025) 12, 124040, arXiv:2504.10130

[12] Peter James Nee et al. (inc. **Lorenzo Pompili**), *Impact of eccentricity and mean anomaly in numerical relativity mergers*

Class. Quant. Grav. 42 (2025) 135011, arXiv:2503.05422

[11] Aldo Gamboa et al. (inc. **Lorenzo Pompili**), *Accurate waveforms for eccentric, aligned-spin binary black holes: The multipolar effective-one-body model SEOBNRv5EHM*

Phys. Rev. D 112 (2025) 4, 044038, arXiv:2412.12823

[10] * **Lorenzo Pompili**, Alessandra Buonanno, Michael Pürrer, *Accounting for Numerical-Relativity Calibration Uncertainty in Gravitational-Wave Modeling and Inference*

Phys. Rev. D 112 (2025) 4, 044062, arXiv:2410.16859

[9] * Félix-Louis Julié, **Lorenzo Pompili**, Alessandra Buonanno, *Inspiral-merger-ringdown waveforms in Einstein-scalar-Gauss-Bonnet gravity within the effective-one-body formalism*

Phys. Rev. D 111 (2025) 2, 024016, arXiv:2406.13654

[8] ** Alessandra Buonanno, Gustav Mogull, Raj Patil, **Lorenzo Pompili**, *Post-Minkowskian Theory Meets the Spinning Effective-One-Body Approach for Bound-Orbit Waveforms*

Phys. Rev. Lett. 133 (2024) 21, 211402, arXiv:2405.19181

[7] Arnab Dhani et al. (inc. **Lorenzo Pompili**), *Systematic Biases in Estimating the Properties of Black Holes Due to Inaccurate Gravitational-Wave Models*

Phys. Rev. X 15 (2025) 3, 031036, arXiv:2404.05811

[6] Alexandre Toubiana, **Lorenzo Pompili** et al., *Measuring source properties and quasi-normal-mode frequencies of heavy massive black-hole binaries with LISA*

Phys. Rev. D 109 (2024) 10, 104019, arXiv:2307.15086

[5] Deyan P. Mihaylov et al. (inc. **Lorenzo Pompili**), *pySEOBNR: a software package for the next generation of effective-one-body multipolar waveform models*

SoftwareX 30 (2025) 102080, arXiv:2303.18203

[4] Maarten van de Meent et al. (inc. **Lorenzo Pompili**), *Enhancing the SEOBNRv5 effective-one-body waveform model with second-order gravitational self-force fluxes*

Phys. Rev. D 108 (2023) 12, 124038, arXiv:2303.18026

[3] Antoni Ramos-Buades et al. (inc. **Lorenzo Pompili**), *SEOBNRv5PHM: Next generation of accurate and efficient multipolar precessing-spin effective-one-body waveforms for binary black holes*

Phys. Rev. D 108 (2023) 12, 124037, arXiv:2303.18046

[2] Mohammed Khalil et al. (inc. **Lorenzo Pompili**), *Theoretical groundwork supporting the precessing-spin two-body dynamics of the effective-one-body waveform models SEOBNRv5*

Phys. Rev. D 108 (2023) 12, 124036, arXiv:2303.18143

[1] * **Lorenzo Pompili** et al., *Laying the foundation of the effective-one-body waveform models SEOBNRv5: improved accuracy and efficiency for spinning non-precessing binary black holes*

Phys. Rev. D 108 (2023) 12, 124035, arXiv:2303.18039

Publications (long author-list with direct personal contribution)

Long author-list publications with direct personal contribution, in reverse chronological order.

As a member of the LIGO Scientific Collaboration, I am a co-author on several full-collaboration papers, see INSPIRE for a full list. Publications from the LIGO-Virgo-KAGRA Collaboration for which I was part of the Editorial Team are marked with *.

[17] * LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GWTC-5.0: Tests of General Relativity*

Submitted to Astrophys. J., arXiv:2607.19293
Contributed to the `pSEOBNR` parametrized ringdown analyses across the catalog events, and served as waveform liaison between the Testing GR and Waveforms working groups.

[16] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog*

Submitted to Astrophys. J. Lett., arXiv:2605.27225
Served on the parameter-estimation rota as an expert member and cross-checked production results with `DINGO`. `SEOBNRv5PHM` was used to measure the parameters of all binary-black-hole signals.

[15] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GW240925 and GW250207: Astrophysical Calibration of Gravitational-wave Detectors*

Phys. Rev. Lett. 137 (2026) 7, 071401, arXiv:2605.11703
Contributed to the `pSEOBNR` ringdown tests of GW250207, including studies of the impact of calibration errors on the results.

[14] Quentin Baghi et al. (inc. **Lorenzo Pompili**), *LISA science ground segment conventions*

arXiv:2603.22377

[13] * LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GWTC-4.0: Tests of General Relativity. III. Tests of the Remnants*

Submitted to Astrophys. J. Lett., arXiv:2603.19021
Led the `pSEOBNR` parametrized ringdown analysis for all catalog events and contributed to the review of the `pyRing` results. Served as liaison between the Testing GR and Parameter Estimation working groups.

[12] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-Spin Black Hole Coalescences*

Astrophys. J. Lett. 993 (2025) 1, L21, arXiv:2510.26931
Measured the eccentricity of both events with `SEOBNRv5EHM` and proposed including these results in the paper, the first LVK paper with a dedicated section on eccentricity measurement.

[11] * LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *Black Hole Spectroscopy and Tests of General Relativity with GW250114*

Phys. Rev. Lett. 136 (2026) 4, 041403, arXiv:2509.08099
Co-proposed the paper and led the `pSEOBNR` analyses, constraining the dominant ringdown frequency to within a few percent of GR and, for the first time, the $\ell = m = 4$ mode. Member of the five-person paper writing team; led the public data release.

[10] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes*

Phys. Rev. Lett. 135 (2025) 11, 111403, arXiv:2509.08054
Performed parameter estimation with the eccentric model `SEOBNRv5EHM`, obtaining the constraint $e \leq 0.03$ reported in the paper.

[9] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GW230814: investigation of a loud gravitational-wave signal observed with a single detector*

Astrophys. J. Lett. 1004 (2026) 2, L23, arXiv:2509.07348
Provided the first `pSEOBNR` results on the event, which helped motivate the proposal of a dedicated paper.

[8] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*

Astrophys. J. Lett. 1004 (2026) 2, L22, arXiv:2508.18082
Served on the parameter-estimation rota, coordinating preliminary analyses of significant events, and reviewed final results as part of the O4a production team. Main point of contact for `SEOBNRv5PHM`, used for all binary-black-hole signals in the catalog.

[7] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *GW231123: a Binary Black Hole Merger with Total Mass* 190–$265\,M_\odot$

Astrophys. J. Lett. 993 (2025) 1, L25, arXiv:2507.08219
Waveform expert in the team performing the initial analyses of the event; contributed to the study and interpretation of the waveform systematics that informed the paper.

[6] Emanuele Berti et al. (inc. **Lorenzo Pompili**), *Black hole spectroscopy: from theory to experiment*

Class. Quant. Grav. 43 (2026) 12, 123001, arXiv:2505.23895

[5] Adrian Abac et al. (inc. **Lorenzo Pompili**), *The Science of the Einstein Telescope*

JCAP 03 (2026) 081, arXiv:2503.12263

[4] Anuradha Gupta et al. (inc. **Lorenzo Pompili**), *Possible Causes of False General Relativity Violations in Gravitational Wave Observations*

SciPost Phys. Comm. Rep. 5 (2025), arXiv:2405.02197

[3] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *Observation of Gravitational Waves from the Coalescence of a* 2.5–$4.5\,M_\odot$ *Compact Object and a Neutron Star*

Astrophys. J. Lett. 970 (2024) 2, L34, arXiv:2404.04248
Performed parameter-estimation analyses of the event with `SEOBNRv5PHM`.

[2] LISA Consortium Waveform Working Group (inc. **Lorenzo Pompili**), *Waveform Modelling for the Laser Interferometer Space Antenna*

Living Rev. Rel. 28 (2025) 1, 9, arXiv:2311.01300

[1] LIGO-Virgo-KAGRA Collaboration (inc. **Lorenzo Pompili**), *Tests of General Relativity with GWTC-3*

Phys. Rev. D 112 (2025) 8, 084080, arXiv:2112.06861
Identified and corrected a bug in the combination of the `pSEOBNR` ringdown results in the original version of the paper.

Talks

Invited talks are marked with *.

09/2026 * gweccentricity Workshop, Corfu – Panel member: *Detecting eccentricity: current evidence and future prospects*

03/2026 LIGO-Virgo-KAGRA Collaboration Meeting, Pisa – *Measuring orbital eccentricity in gravitational-wave events with SEOBNRv6EHM*

02/2026 * Einstein Telescope science workshop for early-career researchers, Rome – *Ready for what lies ahead? Waveform opportunities and challenges towards the Einstein Telescope*

01/2026 * "Lost in Translation" Workshop, virtual – *Effective-one-body waveforms and fluxes in Einstein-scalar-Gauss-Bonnet gravity*

01/2026 GW:UK Workshop, Nottingham – *Testing general relativity with black hole ringdowns*

12/2025 * GWSky Workshop, Potsdam – Panel member: *Machine learning methods for waveform calibration*

12/2025 GWPAW, Atlanta – *Black-hole spectroscopy and tests of general relativity with GW250114*

11/2025 * Astrophysics group seminar, Birmingham – *Testing general relativity with black hole ringdowns*

10/2025 * Self-force group seminar, Southampton – *Ever more accurate effective-one-body waveforms for gravitational-wave astrophysics*

03/2025 LIGO-Virgo-KAGRA Collaboration Meeting, Melbourne – *Incorporating antisymmetric multipoles for spin-precessing binary signals into the SEOBNR framework*

03/2025 * AstroBicocca seminar, Milan – *Systematic biases in gravitational-wave models: incorporating waveform uncertainty in the parameter estimation of binary-black-hole signals*

03/2025 * TianQin student seminar, virtual – *Systematic biases in gravitational-wave models: incorporating waveform uncertainty in the parameter estimation of binary-black-hole signals*

09/2024 LIGO-Virgo-KAGRA Collaboration Meeting, Barcelona – *Accounting for numerical-relativity calibration uncertainty in gravitational-wave modeling and inference*

09/2024 TEONGRAV Workshop, Rome – *Inspiral-merger-ringdown waveforms in Einstein-scalar-Gauss-Bonnet gravity within the effective-one-body formalism*

09/2024 * Fundamental Physics Meets Waveforms With LISA, Potsdam – *SEOBNRv5: advancements in effective-one-body gravitational waveforms towards LISA*

09/2024 * Fundamental Physics Meets Waveforms With LISA, Potsdam – Panel member: *LISA MBHB accuracy requirements*

09/2024 * Fundamental Physics Meets Waveforms With LISA, Potsdam – Panel member: *Impact of systematics on tests of GR*

07/2024 * PAX IX Workshop, London – Panel member: *Waveform challenges and numerical relativity*

03/2024 LIGO-Virgo-KAGRA Collaboration Meeting, Baton Rouge – *Inspiral-merger-ringdown waveforms in Einstein-scalar-Gauss-Bonnet gravity within the effective-one-body formalism*

09/2023 LIGO-Virgo-KAGRA Collaboration Meeting, Toyama – *Accounting for numerical relativity calibration uncertainty in modeling and inference of gravitational waves*

07/2023 26th Capra Meeting on Radiation Reaction in General Relativity, Copenhagen – *Enhancing the SEOBNRv5 effective-one-body waveform model with second-order gravitational self-force fluxes*

09/2022 LIGO-Virgo-KAGRA Collaboration Meeting, Cardiff – *Update on SEOBNR waveform development for O4a*

03/2022 DPG Spring Meeting, Heidelberg (virtual) – *New generation effective-one-body waveforms for binary black holes with non-precessing spins*

Teaching experience

2022 Teaching assistant for Prof. Buonanno's course on Gravitational Waves, *Humboldt University, Berlin*

Mentoring

As lead developer of `pySEOBNR`, I regularly support students and postdocs who use and extend the code. I have also informally mentored the following PhD students more closely:

- 2026– Brian Sheridan, *University of Nottingham* – time–frequency representations for simulation-based inference of gravitational-wave signals with `DINGO`
- 2026– Alexander Roussopoulos, *University of Nottingham* – applications of `DINGO` to parametrized inspiral tests of general relativity
- 2025– Cedric Ewen, *Max Planck Institute for Gravitational Physics and Max Planck Institute for Intelligent Systems* – JAX-based and machine-learning approaches to waveform modeling
- 2025– Leonardo Grimaldi, *Max Planck Institute for Gravitational Physics* – tests of general relativity with `pSEOBNR`, waveform modeling in Einstein-scalar-Gauss-Bonnet gravity, applications of `DINGO` to tests of general relativity
- 2025–2026 Cheng Foo, *Max Planck Institute for Gravitational Physics* – development of effective-one-body waveform models in the SEOBNR framework; resulting paper: arXiv:2611.11449

Service

- 2023– Referee for Physical Review D, Classical and Quantum Gravity, European Physical Journal C
- 09/2024 Local organizing committee, *Fundamental Physics Meets Waveforms With LISA* Workshop, Potsdam
- 2023–2025 Organizer of weekly meetings dedicated to LIGO science at the AEI
- 2021–2025 Organizer of weekly meetings dedicated to effective-one-body waveform modeling at the AEI

Outreach

I contributed to the organization of the following outreach events:

- 09/2025 *Sounds of Silence: Ten Years of Gravitational-Wave Astronomy*, Berlin Lange Nacht der Astronomie 2025
- 11/2024 *Deciphering Black Hole Symphonies: The New World of Gravitational Wave Astronomy*, Berlin Science Week 2024

I wrote the science summary for the following papers from the LIGO-Virgo-KAGRA Collaboration:

* GWTC-5.0: Tests of General Relativity
* Black Hole Spectroscopy and Tests of General Relativity with GW250114

IT Skills

Python, Cython, Mathematica, C/C++; HTCondor, Slurm, Git, LaTeX

Open source software

pySEOBNR: accurate and efficient gravitational-wave models for compact binary coalescences using the effective-one-body approach. **Lead developer and maintainer.**

DINGO: parameter estimation of gravitational-wave data using neural posterior estimation. **Developer.**

I contributed to various public codes used for gravitational-wave data analysis, including LALSuite, bilby, bilby_pipe, bilby_tgr and PESummary